



Disaster Recovery (DR) Across Regions

Ensuring AI Systems Survive Regional Failures

A strategic approach to maintaining business continuity when the unexpected happens

Why DR Matters for AI

Business-Critical Applications

AI systems now power essential functions like fraud detection, healthcare diagnostics, and personalized experiences that directly impact revenue and customer trust.

Multiple Protection Layers

DR ensures business continuity, maintains regulatory compliance, and prevents revenue and trust erosion during failures.

Extended Outage Risk

Regional outages can last **hours or days**, creating significant business disruption when AI systems are unavailable.



Threats That Trigger DR

Understanding the range of threats helps organizations prepare appropriate recovery strategies



Cloud Region Outage

Power failures, networking issues, or natural disasters affecting an entire geographic region



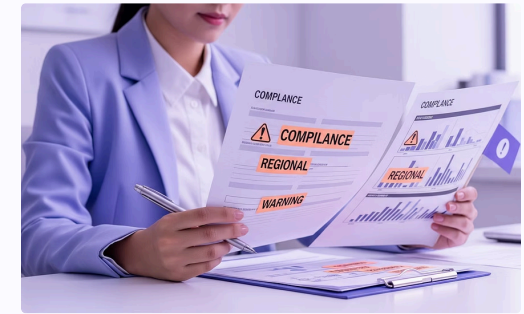
Data Corruption

Ransomware attacks or critical corruption events affecting data integrity



Infrastructure Compromise

Model registry or feature store security breaches requiring isolation



Compliance Issues

Regulatory changes requiring rapid migration between regions

DR Objectives: Setting Clear Recovery Targets

RTO (Recovery Time Objective)

How fast can your AI system recover after disaster?



RPO (Recovery Point Objective)

How much data loss is acceptable during recovery?



- 1 AI System Type
RTO Target
RPO Target

- 2 Financial Fraud Model
< 15 minutes
Near-zero

- 3 Recommendation Engine
< 2 hours
1-2 hours

Single Server
Backup to tape

Mirroed
with Failover

3



4



DR Architectures: Finding Your Recovery Balance

Cold Standby

Cheapest option requiring manual spin-up of resources

- Minimal ongoing costs
- Longest recovery time
- Best for non-critical AI workloads

Warm Standby

Pre-provisioned but scaled-down resources

- Moderate ongoing costs
- Medium recovery time
- Balance of cost vs. readiness

Hot Active-Active

Full replicas operating in multiple regions

- Highest ongoing costs
- Near-immediate recovery
- For mission-critical AI systems

Hybrid

Training in one region, inference replicated

- Optimized cost profile
- Prioritizes customer-facing components
- Balances performance and protection

Data Replication for DR

Ensuring your AI systems can access required data after failover is critical to effective recovery



Object Storage

S3 Cross-Region Replication (CRR), GCS multi-region buckets



Databases

Aurora Global Database, Spanner, Cosmos DB with geo-replication



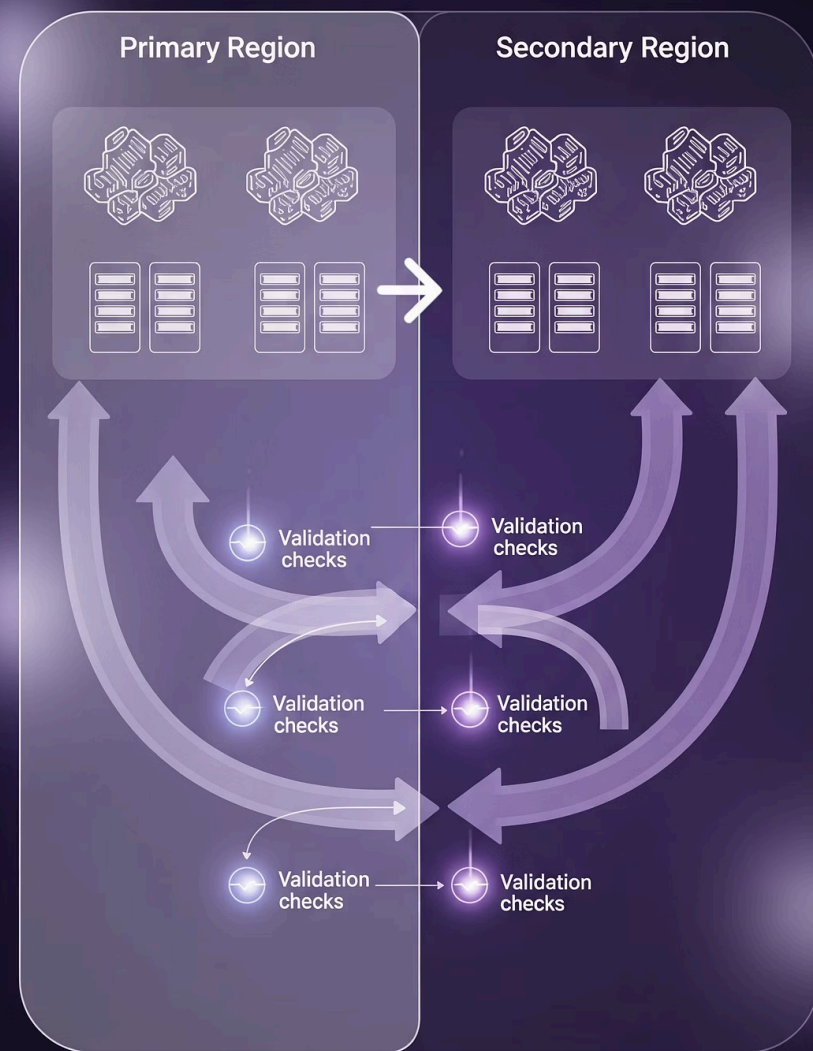
Feature Stores & Vector DBs

Async/sync replicas with configurable consistency levels



Critical Action: Test Failover Regularly

Validate that your RPO/RTO objectives are being met through scheduled replication tests and controlled failover exercises.



Networking & Routing: Seamless Traffic Redirection

1 Global DNS Failover

Implement AWS Route53, Google Cloud DNS, or Azure Traffic Manager to automatically redirect traffic when a region fails

2 Anycast Load Balancing

Route users to the nearest healthy region using global load balancers with health checks

3 Network Redundancy

Ensure VPN connections and cloud interconnects have redundant paths across regions

4 Certificate Management

Verify TLS certificates are valid in all regions to prevent authentication failures during failover



DR for AI Pipelines: Beyond Just Data

Comprehensive DR for AI requires addressing all components in the ML lifecycle



Training Jobs

Implement checkpointing to restart training in another region without losing progress



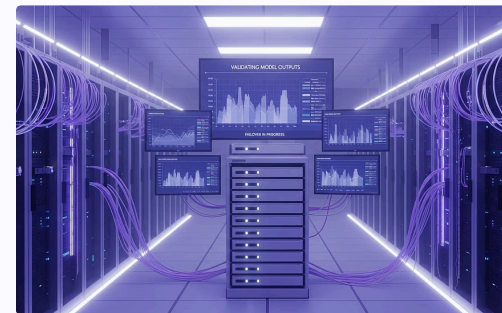
Model Registry

Mirror model artifacts and metadata across clouds/regions for fast deployment



Inference APIs

Deploy multi-region load-balanced endpoints with automated health checks



Shadow Deployments

Use parallel deployments to validate model behavior during region transitions

Testing & Chaos Drills

Hope is not a strategy. Regular testing validates your DR capabilities work when needed.

Regional Outage Simulation

Use chaos engineering tools to safely simulate complete region failures without affecting production

Failover Validation

Verify automatic DNS and service routing actually works as expected during controlled tests

Data Integrity Checks

Confirm replication lag is within RPO and validate data consistency across regions

Team Readiness

Develop and practice recovery playbooks with engineering teams through regular drills



"Everyone has a disaster recovery plan until it's time to execute it."



Compliance & Governance Considerations

Data Residency Constraints

Some regulatory frameworks require **local failover only** within national boundaries, limiting DR options

Certification Requirements

SOC2, ISO 27001, and HIPAA compliance require formal DR documentation and regular testing

Audit Trail Continuity

Maintain uninterrupted audit logs during failover events to preserve compliance records

Data Protection

Ensure data remains encrypted during both replication and recovery processes

 Data residency requirements can significantly impact DR strategy selection and implementation. Work with legal and compliance teams early in the planning process.

Best Practices: Building Resilient AI Infrastructure



Define Clear Recovery Objectives

Establish specific RTO/RPO targets for each AI workload based on business impact



Automate Failover Processes

Reduce human error and recovery time through automated detection and failover procedures



Replicate the Full ML Stack

Ensure models, data, and pipelines are all included in your DR strategy



Regular Testing

Use chaos engineering tools to regularly simulate failures and validate recovery



Cost vs. Risk Balance

Select the appropriate DR tier based on workload criticality and budget constraints

Implementing these practices creates resilient AI systems that can withstand regional failures while maintaining operational continuity.

Key Takeaways

DR is not optional for critical AI systems

As AI becomes business-critical, robust disaster recovery is an essential requirement

Multiple architecture options exist

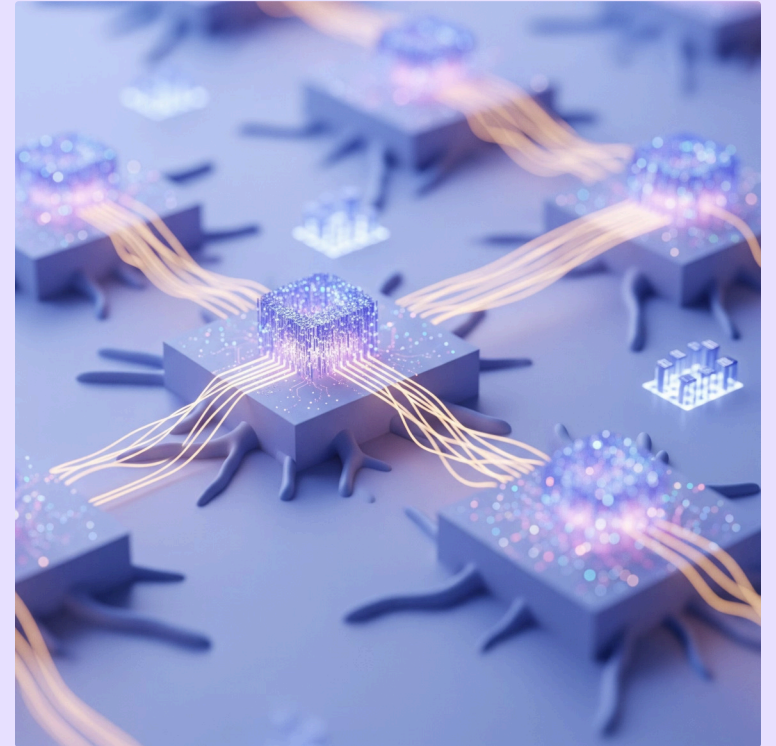
Cold, warm, hot, and hybrid approaches offer flexibility to balance cost vs. recovery speed

Comprehensive strategy required

Cross-region replication, DNS failover, and regular chaos testing create resilient systems

Align with business requirements

DR implementation must address both compliance needs and business service level agreements



Questions?